Part 1: Launch the DEVASC VM

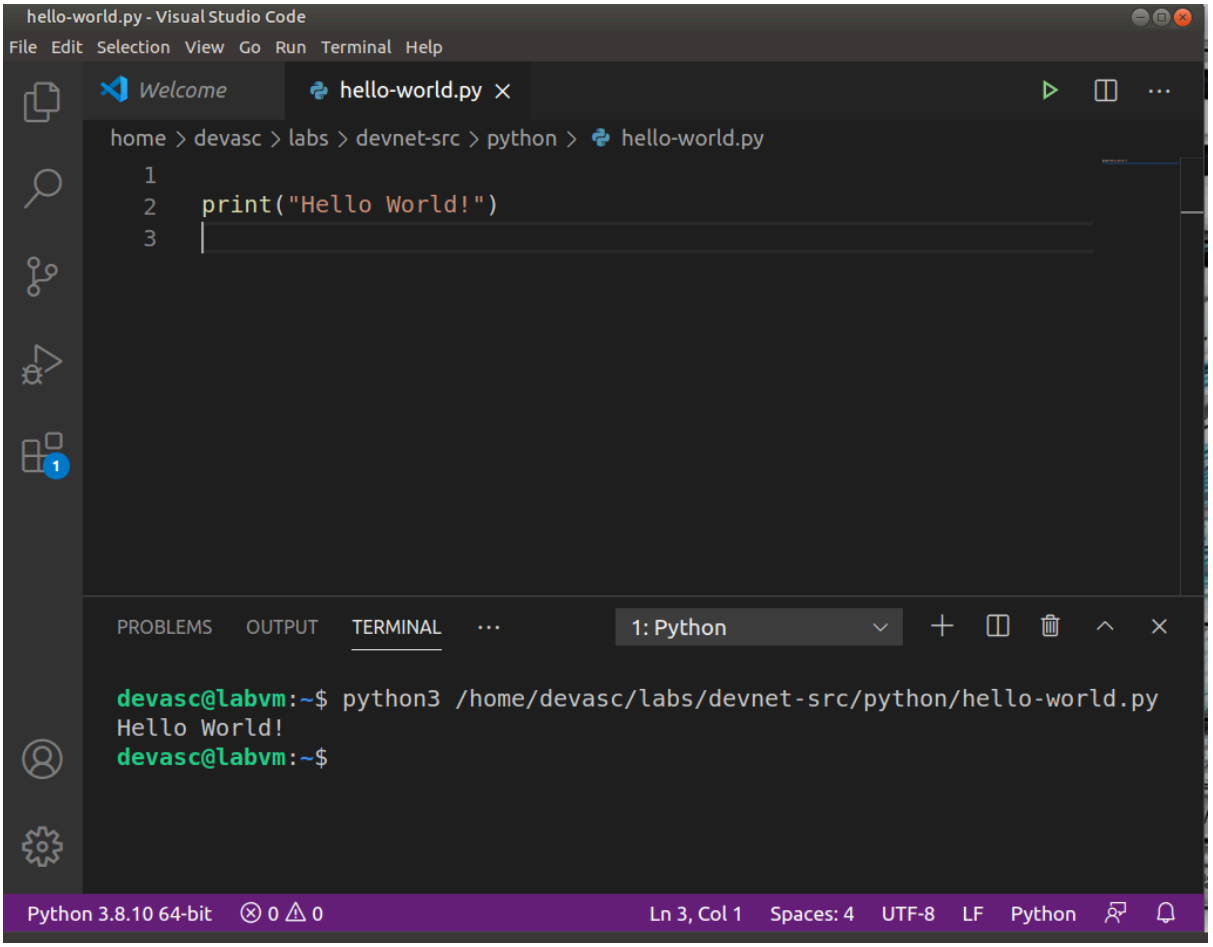
If you have not already completed the Lab - Install the Virtual Machine Lab Environment, do so now. If you

have already completed that lab, launch the DEVASC VM.

Part 2: Starting Python and VS Code

```
devasc@labvm:~$ python3 -V
Python 3.8.10
devasc@labvm:~$ python3
Python 3.8.10 (default, Mar 18 2025, 20:04:55)
[GCC 9.4.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> 2 + 3
5
>>> 10 - 4
6
>>> 2 * 4
8
>>> 20 / 5
4.0
>>> 3 ** 2
9
>>> print("Hello World!")
Hello World!
>>> quit()
devasc@labvm:~$
```

Step 4: Open VS Code and create a script for Hello World.



```
File  Edit  View  Search  Terminal  Help
devasc@labvm:~$ python3 ~/labs/devnet-src/python/hello-world.py
Hello World!
devasc@labvm:~$
```

Part 3: Review Data Types and Variables

In this part, you will use the interactive interpreter to review data types, create variables, concatenate strings,
and cast between data types.

Step 1: Use the interactive interpreter to review basic data types.

```
devasc@labvm:~$ python3
Python 3.8.10 (default, Mar 18 2025, 20:04:55)
[GCC 9.4.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> type(98)
<class 'int'>
>>> type(98.6)
<class 'float'>
>>> type("Hi!")
<class 'str'>
>>> type(True)
<class 'bool'>
>>> x = 3
>>> print(x * 5)
15
>>> print("Cisco" * x)
CiscoCiscoCisco
>>>
>>> str1 = "Cisco"
>>> str2 = "Networking"
>>> str3 = "Academy"
>>> print(str1, str2, str3)
Cisco Networking Academy
>>>
```

```
>>> print("The value of x is " + x)
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
TypeError: can only concatenate str (not "int") to str
>>> print("The value of x is " + str(x))
The value of x is 3
>>> type(x)
<class 'int'>
>>> x = str(x)
>>> type(x)
<class 'str'>
>>> num = 22/7
>>> f"The value of num is {num}"
'The value of num is 3.142857142857143'
>>> pi = "{:.2f}".format(num)
>>> f"The value of pi is {pi}."
'The value of pi is 3.14.'
```

```
>>> x = 3
>>> x * 5
15
>>> 1<=1
True
>>> 1>=1
True
>>> "Cisco" * x
'CiscoCiscoCisco'
>>>
```

Part 4: Review Lists and Dictionaries

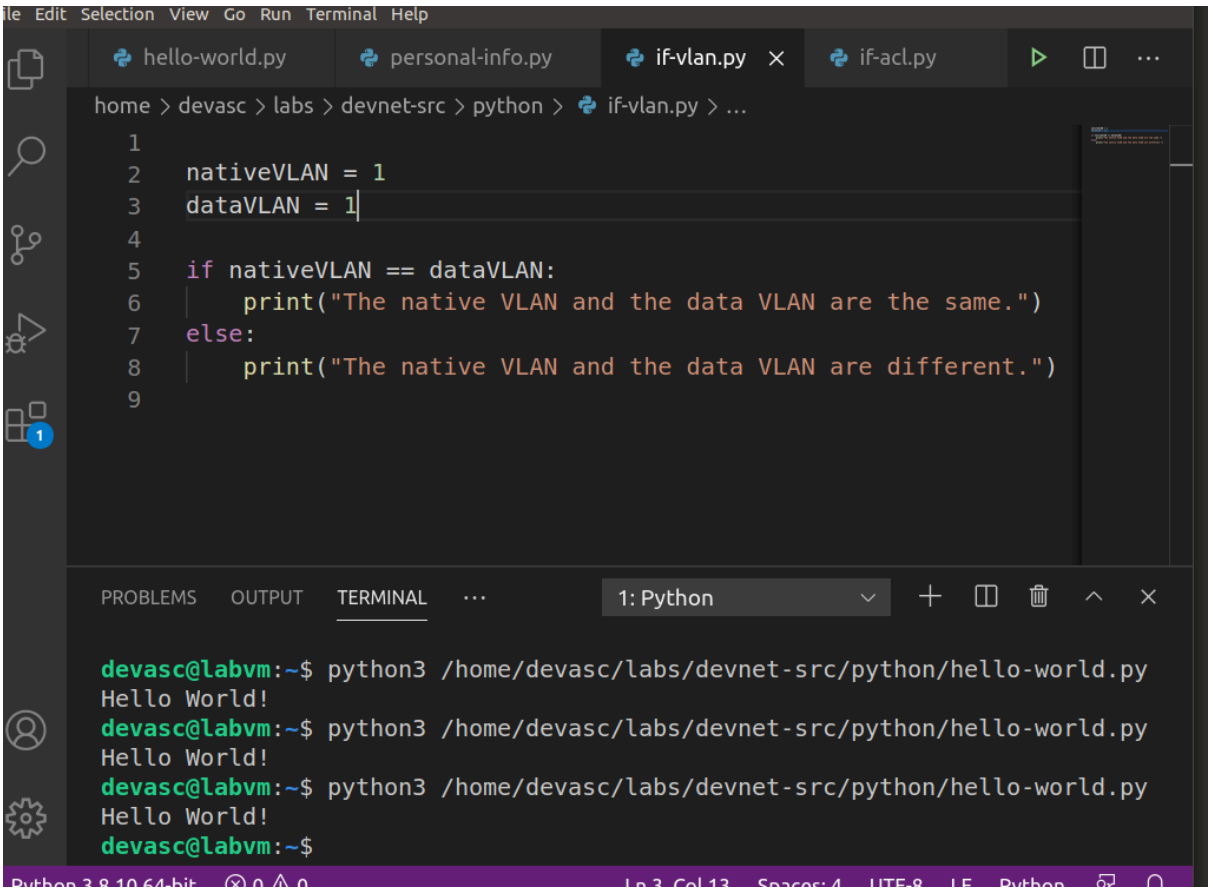
```
>>> hostnames = ["R1", "R2", "R3", "S1", "S2"]
>>> type(hostnames)
<class 'list'>
>>> len(hostnames)
5
>>> hostnames
['R1', 'R2', 'R3', 'S1', 'S2']
>>> hostnames[0]
'R1'
>>> hostnames[-1]
'S2'
>>> hostnames[0] = "RTR1"
>>> hostnames
['RTR1', 'R2', 'R3', 'S1', 'S2']
>>> ipAddress = {"R1": "10.1.1.1", "R2": "10.2.2.1", "R3": "10.3.3.1"}
>>> type(ipAddress)
<class 'dict'>
>>> ipAddress
{'R1': '10.1.1.1', 'R2': '10.2.2.1', 'R3': '10.3.3.1'}
>>> ipAddress["R1"]
'10.1.1.1'
>>> ipAddress["S1"] = "10.1.1.10"
>>> ipAddress
{'R1': '10.1.1.1', 'R2': '10.2.2.1', 'R3': '10.3.3.1', 'S1': '10.1.1.10'}
>>> ipAddress["R3"] = ["10.3.3.1", "10.3.3.2", "10.3.3.3"]
>>> ipAddress
{'R1': '10.1.1.1', 'R2': '10.2.2.1', 'R3': ['10.3.3.1', '10.3.3.2', '10.3.3.3'], 'S1': '10.1.1.10'}
>>>
```

```
1
2   firstName = input("What is your first name? ")
3   lastName = input("What is your last name? ")
4   location = input("What is your location? ")
5   age = input("What is your age? ")
6
7   print("Hi " + firstName + " " + lastName + "! Your location is
8
```

```
>>> firstName = input("What is your first name? ")
What is your first name?
>>> print("Hello " + firstName + "!")
Hello !
>>> firstName = input("What is your first name? ")
What is your first name? Bekmuraz
>>> print("Hello " + firstName + "!")
Hello Bekmuraz!
>>> exit()
devasc@labvm:~$ cd ~/labs/devnet-src/python && code personal-info.py
devasc@labvm:~/labs/devnet-src/python$ cd ~/labs/devnet-src/python && python3 personal-info.py
What is your first name? Bob
What is your last name? Smith
What is your location? London
What is your age? 14
Hi Bob Smith! Your location is London and you are 14 years old.
devasc@labvm:~/labs/devnet-src/python$
Menu  personal-info.py - Visu...  devasc@labvm: ~/labs...
```

Part 6: Review If, For, and While Functions

```
devasc@labvm:~/labs/devnet-src/python$ cd ~/labs/devnet-src/python
devasc@labvm:~/labs/devnet-src/python$ code if-vlan.py
devasc@labvm:~/labs/devnet-src/python$ python3 if-vlan.py
The native VLAN and the data VLAN are different.
devasc@labvm:~/labs/devnet-src/python$ python3 if-vlan.py
The native VLAN and the data VLAN are the same.
devasc@labvm:~/labs/devnet-src/python$ code if-acl.py
devasc@labvm:~/labs/devnet-src/python$ python3 if-acl.py
What is the IPv4 ACL number? 10
This is a standard IPv4 ACL.
devasc@labvm:~/labs/devnet-src/python$ python3 if-acl.py
What is the IPv4 ACL number? 110
This is an extended IPv4 ACL.
devasc@labvm:~/labs/devnet-src/python$ python3 if-acl.py
What is the IPv4 ACL number? 1212
This is not a standard or extended IPv4 ACL.
devasc@labvm:~/labs/devnet-src/python$
Menu  if-acl.py - Visual Studio ...  devasc@labvm: ~/labs...
```




```
1
2   aclNum = int(input("What is the IPv4 ACL number? "))
3
4   if aclNum >= 1 and aclNum <= 99:
5       print("This is a standard IPv4 ACL.")
6   elif aclNum >= 100 and aclNum <= 199:
7       print("This is an extended IPv4 ACL.")
8   else:
9       print("This is not a standard or extended IPv4 ACL.")
10
```

```
devasc@labvm:~/labs/devnet-src/python$ python3
Python 3.8.10 (default, Mar 18 2025, 20:04:55)
[GCC 9.4.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> devices = ["R1", "R2", "R3", "S1", "S2"]
>>> for item in devices:
...     print(item)
...
R1
R2
R3
S1
S2
>>> for item in devices:
...     if "R" in item:
...         print(item)
...
R1
R2
R3
>>> switches = []
>>> for item in devices:
...     if "S" in item:
...         switches.append(item)
...
>>>
>>> for item in devices:
...     print(item)
... switches = []
  File "<stdin>", line 3
    switches = []
    ^
SyntaxError: invalid syntax
>>> for item in devices:
...     if "S" in item:
...         switches.append(item)
...
>>> switches
['S1', 'S2', 'S1', 'S2']
>>>
```

Step 4: While Loop

```
devasc@labvm:~/labs/devnet-src/python$ cd ~/labs/devnet-src/python
devasc@labvm:~/labs/devnet-src/python$ code while-loop.py
devasc@labvm:~/labs/devnet-src/python$ python3 while-loop.py
Enter a number to count to: 10
1
2
3
4
5
6
7
8
9
10
devasc@labvm:~/labs/devnet-src/python$ python3 while-loop.py
Enter a number to count to: 10
1
2
3
4
5
6
7
8
9
10
devasc@labvm:~/labs/devnet-src/python$
```

Menu | while-loop.py - Visual S... | devasc@labvm: ~/labs...

home > devasc > labs > devnet-src > python > while-loop.py > ...

```
1
2   x = input("Enter a number to count to: ")
3   x = int(x)
4
5   y = 1
6
7   while True:
8       print(y)
9       y = y + 1
10      if y > x:
11          break
12
```

```
devasc@labvm:~/labs/devnet-src/python$ python3 while-loop.py
Enter a number to count to: 10
1
2
3
4
5
6
7
8
9
10
Enter a number to count to: 3
1
2
3
Enter a number to count to: 5
1
2
3
4
5
Enter a number to count to: quit
devasc@labvm:~/labs/devnet-src/python$
```

```
1
2   while True:
3       x = input("Enter a number to count to: ")
4
5       if x == 'q' or x == 'quit':
6           break
7
8       x = int(x)
9       y = 1
10
11      while True:
12          print(y)
13          y = y + 1
14          if y > x:
15              break
16
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

Part 7: Review Methods for File Access




```
devasc@labvm:~/labs/devnet-src/python$ cd ~/labs/devnet-src/python
devasc@labvm:~/labs/devnet-src/python$ code file-access.py
devasc@labvm:~/labs/devnet-src/python$ python3 file-access.py
Cisco 819 Router

Cisco 881 Router

Cisco 888 Router

Cisco 1100 Router

Cisco 4321 Router

Cisco 4331 Router

Cisco 4351 Router

Cisco 2960 Catalyst Switch

Cisco 3850 Catalyst Switch

Cisco 7700 Nexus Switch

Cisco Meraki MS220-8 Cloud Managed Switch

Cisco Meraki MX64W Security Appliance

Cisco Meraki MX84 Security Appliance

Cisco Meraki MC74 VoIP Phone

Cisco 3860 Catalyst Switch

devasc@labvm:~/labs/devnet-src/python$
```

```python

file = open("devices.txt", "r")

for item in file:
    item = item.strip()
    print(item)

file.close()

```



home > devasc > labs > devnet-src > python > file-access.py > ...

```python

devices = []

file = open("devices.txt", "r")

for item in file:
    item = item.strip()
    devices.append(item)

file.close()

print(devices)

```

```
devasc@labvm:~/labs/devnet-src/python$ python3 file-access.py
['Cisco 819 Router', 'Cisco 881 Router', 'Cisco 888 Router', 'Cisco 1100 Router', 'Cisco 4321 Router', 'Cisco 4331 Router', 'Cisco 4351 Router', 'Cisco 2960 Catalyst Switch', 'Cisco 3850 Catalyst Switch', 'Cisco 7700 Nexus Switch', 'Cisco Meraki MS220-8 Cloud Managed Switch', 'Cisco Meraki MX64W Security Appliance', 'Cisco Meraki MX84 Security Appliance', 'Cisco Meraki MC74 VoIP Phone', 'Cisco 3860 Catalyst Switch']
devasc@labvm:~/labs/devnet-src/python$
```

```
1
2   file = open("devices.txt", "a")
3
4   while True:
5       newItem = input("Enter device name: ")
6
7       if newItem == "exit":
8           print("All done!")
9           break
10
11      file.write(newItem + "\n")
12
13  file.close()
```

PROBLEMS OUTPUT TERMINAL ··· 1: Python

```
devasc@labvm:~/labs/devnet-src/python$ code file-access-input.py
devasc@labvm:~/labs/devnet-src/python$ python3 file-access-input.py
Enter device name: Cisco 1941 Router
Enter device name: Cisco 2950 Catalyst Switch
Enter device name: Exit
Enter device name: exit
All done!
devasc@labvm:~/labs/devnet-src/python$
```